

When to Use

To test if two or more populations or subpopulations have different patterns of response to a qualitative/categorical DV. This can also be characterized as a test of a pattern of relationship between two qualitative/categorical variables.

H Research Hypothesis: The researcher hypothesized that films and serials would tend to have larger age gaps between the actors playing Lois Lane and Clark Kent (Average or Big), whereas TV shows would tend to have minimal age gaps.

H Null Hypothesis: There is no pattern of relationship between media type (Film/Serial vs TV Show) and age gap category (Minimal vs Average/Big) in the population of Superman portrayals represented by this sample.

> Running a Chi-Square Test

Analyze → Descriptive Statistics → Crosstabs

- Highlight the variable you want to define the rows (be sure it is qualitative/categorical) and click arrow
- Highlight the variable you want to define the columns (be sure it is qualitative/categorical) and click arrow
- “Statistics” — check that you want a “Chi-square analysis”

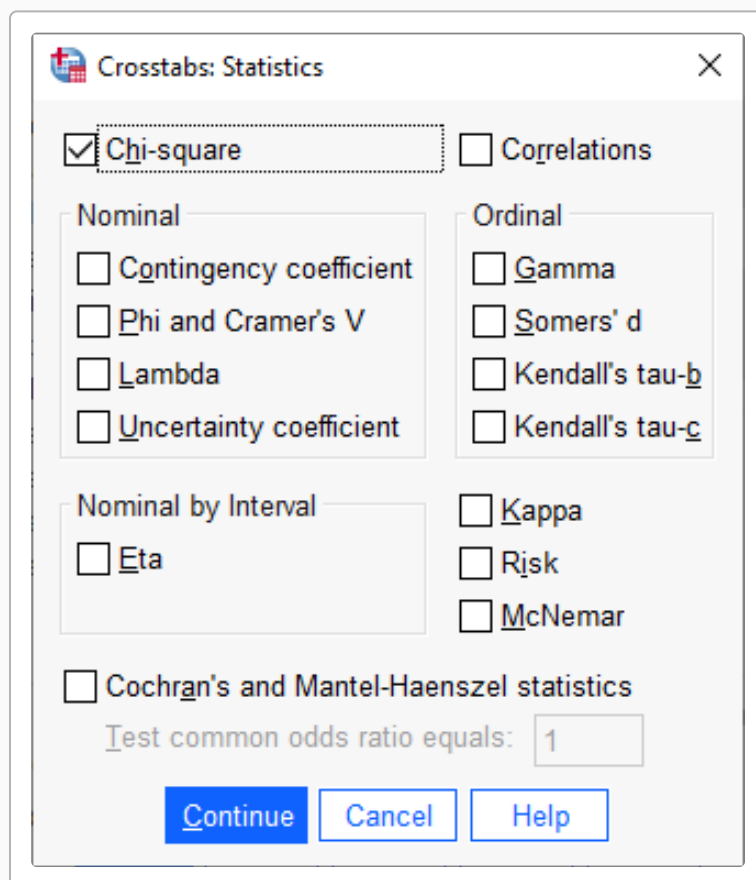
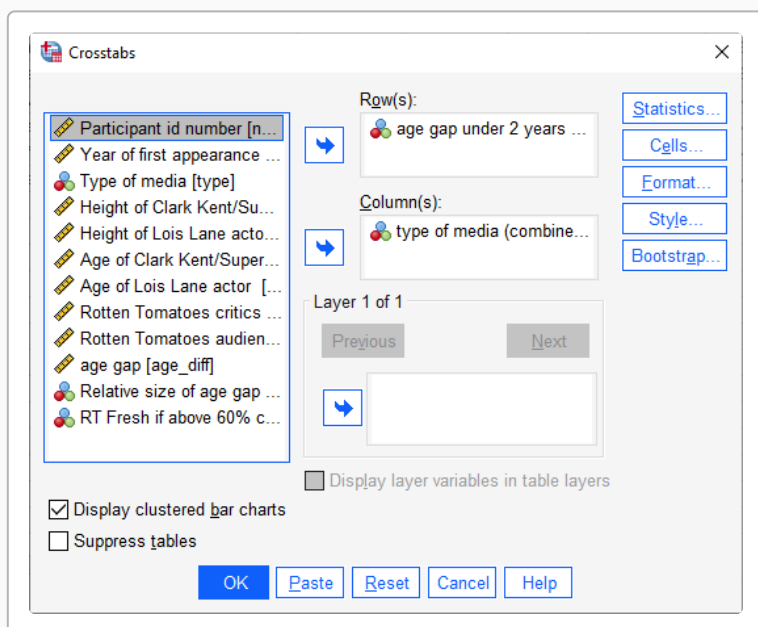
SPSS Syntax

CROSSTABS

```
/TABLES=type2 BY age_grp2
/STATISTICS=CHISQ
/CELLS=COUNT ROW COLUMN TOTAL.
```

①
②
③

- ① Use "row variable" BY "column variable"
- ② Get Chi-square significance test
- ③ Get various row, column and/or total cell percentages (optional)



SPSS Output

Chi-Square Tests

	Value	df	Asymptotic Significance (2- sided)	Exact Sig. (2- sided)	Exact Sig. (1- sided)
Pearson Chi-Square	5.143	1	.023		
Continuity Correction	2.009	1	.156		
Fisher's Exact Test				.083	.083
N of Valid Cases	9				

4 cells (100.0%) have expected count less than 5. The minimum expected count is 0.67. Continuity correction and Fisher's exact test are computed only for a 2x2 table. Because at least one expected count is below 5, use Fisher's exact two-sided p-value for inference.

type2 * age_grp2 Crosstabulation

type2		Average/Big	Minimal	Total
Film/Serial	Count	6	0	6
	Expected Count	4.7	1.3	6.0
TV Show	Count	1	2	3
	Expected Count	2.3	0.7	3.0
Total	Count	7	2	9
	Expected Count	7.0	2.0	9.0

Assuming the null hypothesis is true, a p-value is the probability of obtaining data at least as inconsistent with the null hypothesis as the observed data. It is not the probability that the null hypothesis is true or that this result is a Type I error.

SPSS may display **.000** when a very small p-value rounds to three decimal places. A p-value is not exactly zero; report **p < .001**.

All four expected counts in this example are below 5. Therefore, the Pearson chi-square approximation and the continuity-corrected result are shown for comparison, but Fisher's exact two-sided test is the authoritative inference.

For a 2x2 table, use Fisher's exact test when any expected count is below 5. When all expected counts are at least 5, use Pearson's chi-square as the primary test.

Reporting the Results

It is important to show the cell and marginal frequencies (often also percentages) in a contingency table before presenting the Chi-square results.

As in the example, be sure to communicate:

- The research hypothesis (if there is one)
- The statistical results
- Whether or not those results support the research hypothesis

Table 1

Crosstabulation of Media Type and Age Gap Category

Media Type	Age Gap Category		
	Average/Big	Minimal	Total
Film/Serial	6 (100%)	0 (0%)	6 (100%)
TV Show	1 (33.3%)	2 (66.7%)	3 (100%)
Total	7	2	9

Note. Fisher's exact test (two-sided), $p = .083$.

Example Write-Up

¶ Sample APA Write-Up

Table 1 shows the contingency table for these variables. The sample of Superman portrayals was divided between the two types of media (6 Film/Serial and 3 TV Show) and between the two age-gap categories (2 Minimal and 7 Average/Big). Inspection of the table suggests that, as hypothesized, films and serials tended to have larger age gaps, whereas TV shows tended to have minimal age gaps. Because every expected count was below 5, Fisher's exact test was used for inference. Contrary to the hypothesis, the relationship was not statistically significant, Fisher's exact test (two-sided), $p = .083$.

i Note

This example shows how to structure a write-up following the guidelines: stating the hypothesis, describing the table, reporting the chi-square results, and interpreting the findings in relation to the hypothesis.